

AB RENEWABLE ENERGY ENGINEERING

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WINDPUMP

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1. Power generated from moving air.

- a. Wind power
- b. Hydropower
- c. Photovoltaic cell
- d. None of the above

2. Wind is moving at 4 meters per second. If the rotor area is 2 m², what is the wind power?

- a. 80 watts
- b. 125 watts
- c. 95 watts
- d. None of the above

$$\frac{1}{2}(1.25)(2)(4)^3$$

3. Device that converts the kinetic energy of the wind to mechanical power.

- a. Rotor
- b. Generator
- c. Motor
- d. None of the above

4. The greater the number of blades of the wind machine, _____ the torque.

- a. the higher
- b. the lower
- c. the same
- d. None of the above

5. Wind machine used for water pumping with axis rotation perpendicular to the wind direction.

- a. Horizontal-axis rotor windpump
- b. Vertical-axis rotor windpump
- c. Cross-flow rotor windpump
- d. None of the above

6. The wind moves on a 2m-diameter rotor at a rate of 2 m/s. It suddenly moves at 4 m/s, what would be the power output of the windmill?

- a. 4 times
- b. 6 times
- c. 8 times
- d. None of the above

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7. A windpump is to be installed in an area where ground water has lots of silt. As an Agricultural Engineer, which of the following courses of action would you recommend in setting up the wind pump?

- a. Use a 4in. pump casing for the suction pipe of the wind pump.
- b. Use a 4in. pump casing with fine screen mesh for the suction pipe of the wind pump.
- c. Use concrete dug-well casing for the suction pipe of the windpump.
- d. None of the above

8. Windpumps are suitable for high _____ operation.

- a. torque
- b. speed — *wind turbine*
- c. power
- d. None of the above

9. Doubling the diameter of the horizontal-axis rotor of a wind machine will increase its power by _____.

- a. two times
- b. four times
- c. eight times
- d. None of the above

10. Doubling the speed of the wind blowing the horizontal-axis rotor of a wind machine will increase its power by _____ times.

- a. two
- b. four
- c. eight
- d. None of the above

11. Wind machine used for lifting water.

- a. Aerogenerator
- b. Wind turbine
- c. Wind pump
- d. None of the above

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12. Speed of the horizontal-axis rotor wind pump is limited up to ____ rpm.

- a. 20
- ☒ b. 30
- c. 40
- d. None of the above

13. Part of the wind pump that directs the rotor to face perpendicularly with the wind direction.

- a. Head Assembly
- b. Tower Assembly
- ☒ c. Tail-Vane Assembly
- d. All of the above
- e. None of the above

14. For a wind pump that is going to lift water from a 40ft well, it is recommended to install the pump ____.

- a. at the ground surface
- ☒ b. at the well near the level of water

- c. at a level 1 meter from the ground surface
- d. All of the above
- e. None of the above

15. Wind machine used for milling grains.

- a. Windpump
- ☒ b. Windmill
- c. Wind turbine
- d. None of the above

16. If the rotor of a windmill originally at 2 meters is to be reduced to 1 meter, the power output of the machine will be reduced by ____ times.

- a. two
- b. three
- ☒ c. four
- d. None of the above

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17. In an area surrounded by trees, it is recommended that the wind pump rotor be installed ____ the trees.

- a. at the same height with
- ☒ b. above
- c. below
- d. All of the above
- e. None of the above

18. Windpump used for deep-water pumping is usually provided with ____.

- a. direct-drive transmission
- ☒ b. gear-reduction drive
- c. chain-reduction drive
- d. None of the above

19. Area of the rotor surface facing the wind to the total swept area of the rotor.

- ☒ a. Solidity
- b. Angle of swept
- c. Rotor swept
- d. None of the above

20. Ratio of the mechanical power actually attained directly from the rotor and from the wind power.

- a. Torque coefficient
- ☒ b. Power coefficient
- c. Rotor coefficient
- d. None of the above

21. Wind machine viability refers to ____.

- a. the available wind to operate the machine
- b. life span of the wind machine in performing the job
- ☒ c. cost of the competing energy for wind machines
- d. None of the above

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GOD BLESS !!!

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